



Questioning the Coverage–Length Metric in Conformal Prediction:

When Shorter Intervals Are Not Better

Yizhou Min¹ Yizhou Lu² Lanqi Li¹ Zhen Zhang¹ Jiaye Teng^{1†}

¹Shanghai University of Finance and Economics ²Fudan University

^{*}equal contribution [†]corresponding author



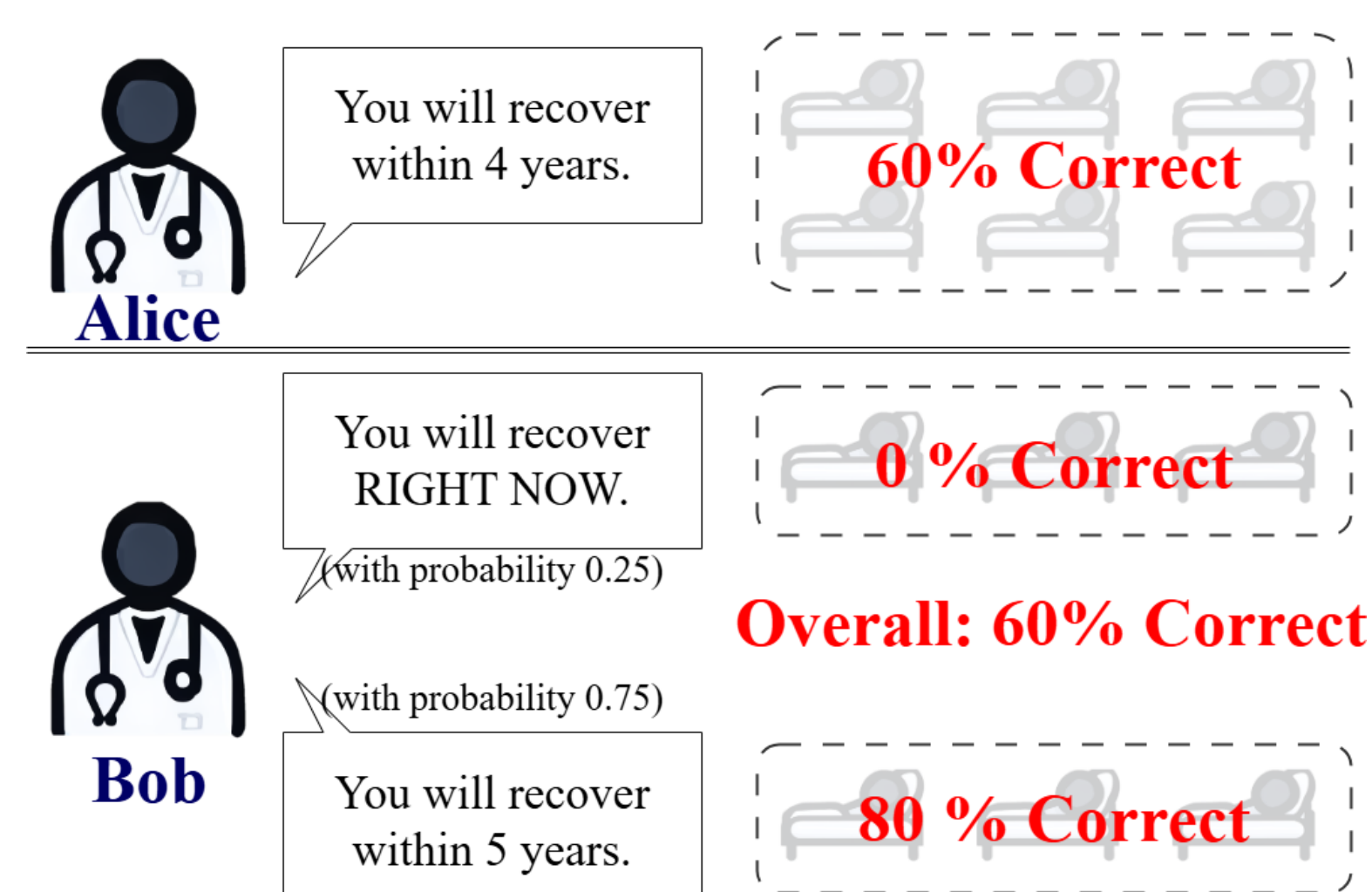
Coverage + shorter average length $\neq$ a better conformal predictor: a randomized wrapper keeps coverage valid yet shortens reported length via *empty* sets — **Interval Stability** exposes it.

1 Is coverage–length the whole story?

Conformal prediction (CP) returns a set $\mathcal{C}_{1-\alpha}(X)$ with $\mathbb{P}\{Y \in \mathcal{C}_{1-\alpha}(X)\} \geq 1 - \alpha$; methods are ranked by **coverage** and **average interval length**.

Key question. If a new algorithm outperforms existing ones on coverage and length, should it automatically be considered superior for practical deployment?

2 A motivating example: Bob wins the metrics



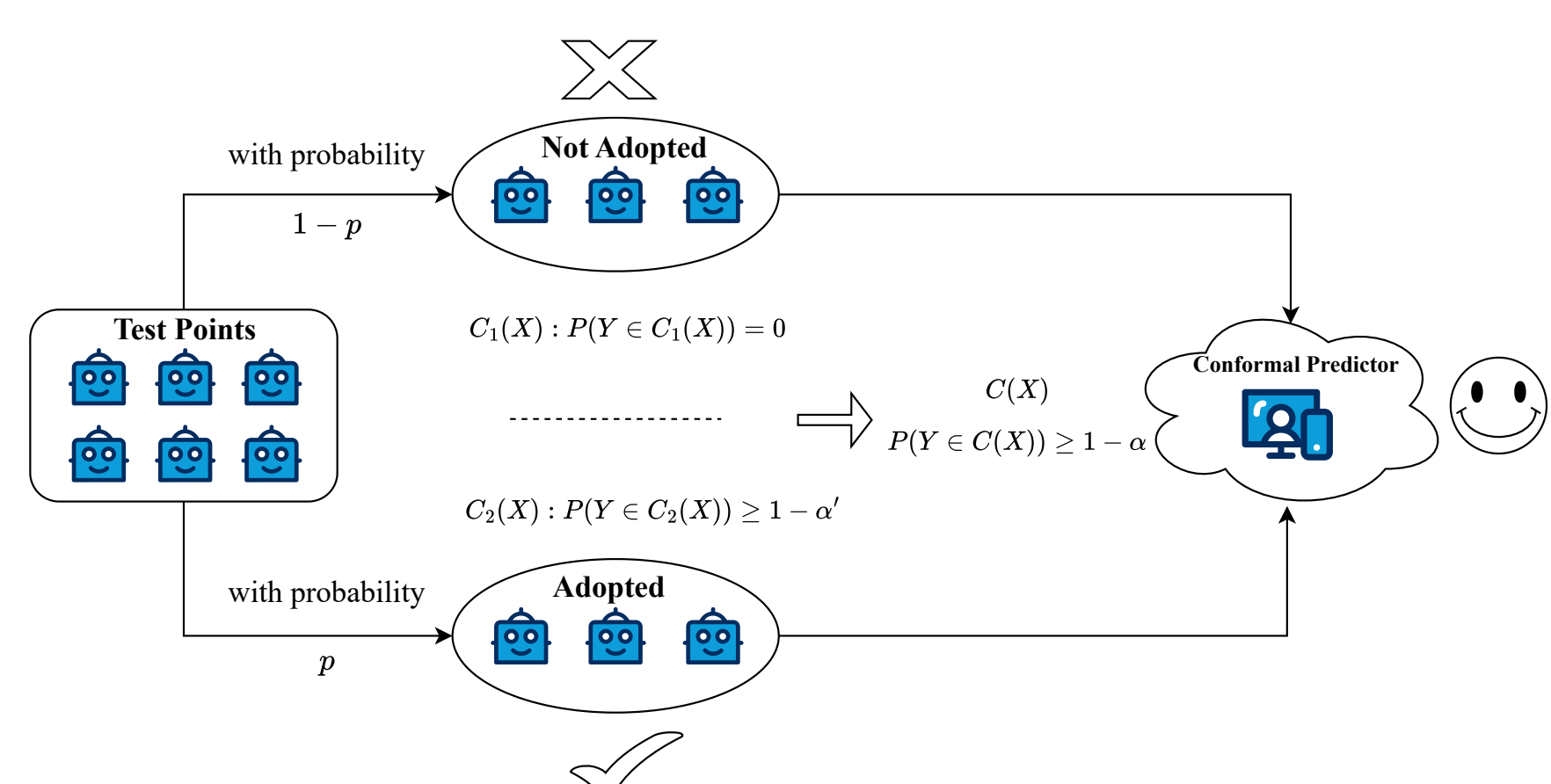
Alice: always $[0, 4]$ years — coverage 60%, length 4. Bob: $[0, 5]$ w.p. 0.75, $[0, 0]$ w.p. 0.25 — coverage 60%, length **3.75**.

Bob wins both metrics, yet 25% of his patients get a vacuous answer and repeated queries disagree.

3 The Prejudicial Trick (PT)

Pick $p \in (1 - \alpha, 1)$ and the more conservative level $\alpha' = 1 - \frac{1-\alpha}{p} < \alpha$:

$$\mathcal{C}_{1-\alpha}^{\text{PT}}(x) = \begin{cases} \text{null set,} & \text{w.p. } 1 - p, \\ \mathcal{C}_{1-\alpha'}^{\text{CP}}(x), & \text{w.p. } p. \end{cases}$$



PT wraps any base CP algorithm (VCP, CQR, RAPS, ...); mixing restores $p(1 - \alpha') = 1 - \alpha$.

PT is a **cautionary counterexample**, not a method: better averages, but unstable and unfair for individuals.

4 Theory: the metric is provably hackable

Coverage cannot flag PT. Under exchangeability, PT keeps marginal coverage $\geq 1 - \alpha$; if the base CP is conditionally valid, PT *inherits* conditional coverage, since its randomness is independent of x .

Length can improve. Let $\mathcal{G}(u) = \mathbb{E}_X[\mathcal{C}_u^{\text{CP}}(X)]$ be the base length curve, $t = 1 - \alpha$:

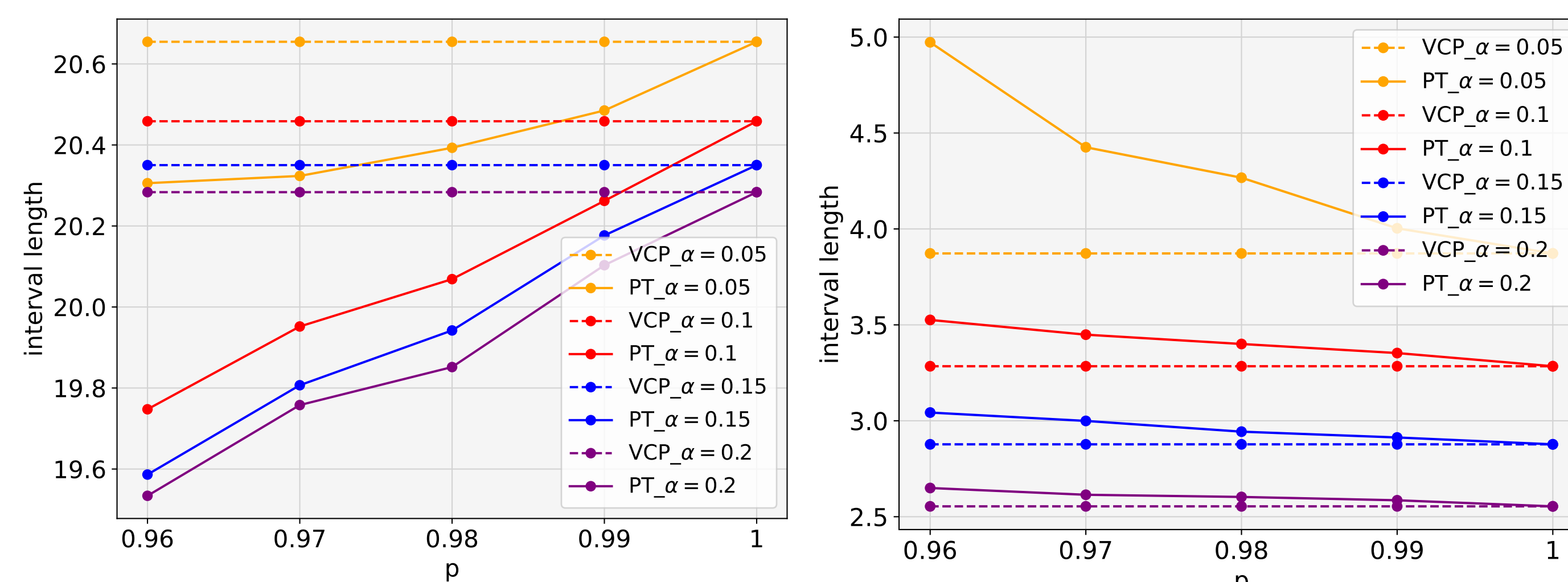
$$\mathbb{E}[\mathcal{C}_t^{\text{PT}}(X)] = p\mathcal{G}(t/p) \implies \text{PT is shorter iff } p\mathcal{G}(t/p) < \mathcal{G}(t),$$

with first-order sufficient condition (and a secant analogue if $\mathcal{G}$ is non-differentiable)

$$\frac{\mathcal{G}(t)}{t} > \mathcal{G}'(t).$$

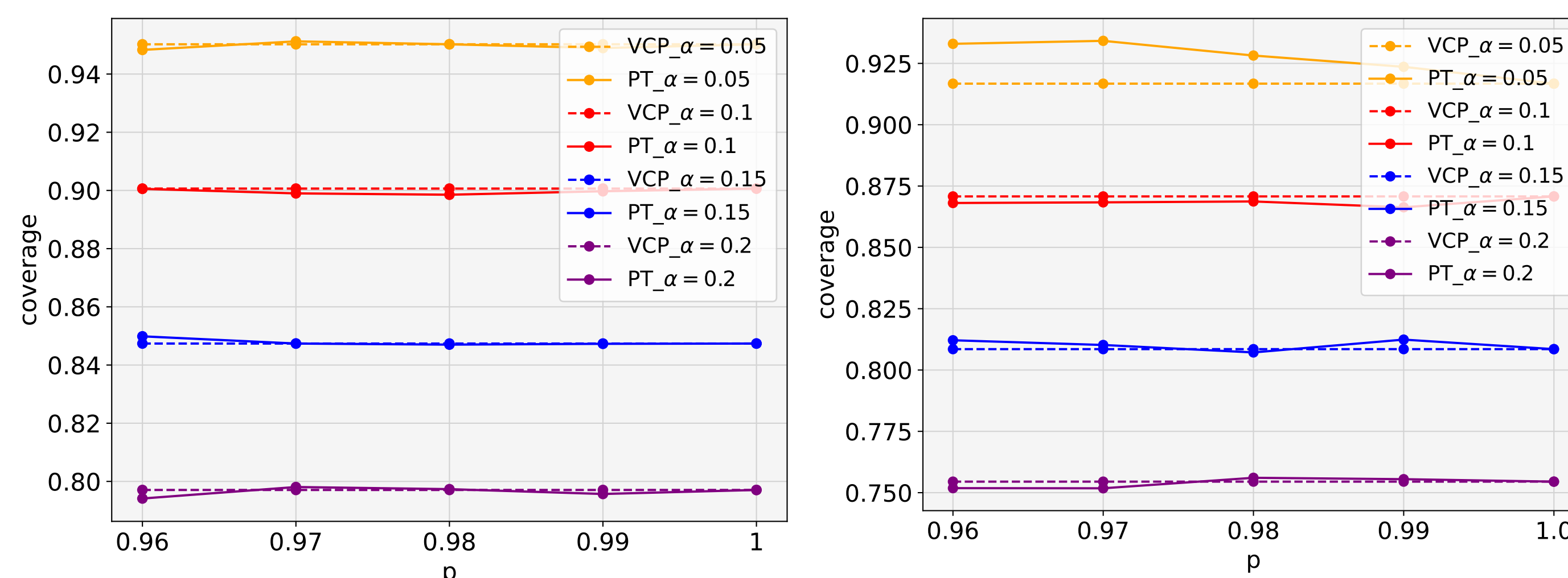
Reading: unless conservative intervals expand super-linearly with coverage, mixing in null sets lowers the average. **Model misspecification** (biased residuals) flattens $\mathcal{G}$ near the target level, so the condition is common in practice — but not universal: for Gaussian scores PT is provably longer for every p .

5 The length gain is real — but conditional



Left: on BIKE (misspecified model) the reported length of PT (solid) drops below VCP (dashed) as p decreases. Right: with Gaussian scores PT is longer for every p — exactly as the theory predicts.

6 Standard coverage audits pass



On BIKE, PT tracks VCP in marginal (left) and month-group (right) coverage for all p and α : coverage audits do not reveal the trick.

7 PT fools the metric across tasks and baselines

Regression ($\alpha=0.1$)	VCP	PT-VCP	Coverage
MEPS-19	42.34	41.92	0.90 / 0.90
BIKE	20.46	19.59	0.90 / 0.90
BIO	21.13	20.44	0.90 / 0.90
CONCRETE	10.32	9.87	0.90 / 0.89
STAR	10.14	9.63	0.91 / 0.91
FACEBOOK-2	20.97	21.01	0.90 / 0.90
ImageNet ($\alpha=0.1$)	RAPS	PT-RAPS	Coverage
ResNet-18	304.0	295.6	0.90 / 0.90
ResNet-50	302.1	290.3	0.90 / 0.90
ResNet-152	301.5	289.0	0.89 / 0.90

Shorter on 9/10 regression datasets and all ImageNet models; same for CQR (10/10), a relaxed PT with small non-null sets, and 7/10 p -value criteria.

8 Interval Stability: make randomness visible

Run-to-run variance of the returned set size on the same input:

$$\text{IS}(\mathcal{C}) = \mathbb{E}_X[\text{Var}_{\mathcal{A}|X, \mathcal{D}_{\text{ex}}}(|\mathcal{C}_{1-\alpha}(X)|)], \quad \text{IS}(\mathcal{C}^{\text{PT}}) = p(1 - p) \mathbb{E}\left[\mathcal{L}\left(x, \frac{1-\alpha}{p}\right)\right]^2 > 0.$$

Method	IS range	Reading
VCP	all 0.00	deterministic: passes
PT-VCP	0.58 – 1.26	vacuous randomness flagged
Localized CP	0.01 – 1.41	retrained scale varies per run
RAPS	0.01 – 0.19	mild tie-breaking randomness
PT-RAPS	11.9 – 12.1	flagged: $\sim 60\times$ above RAPS

IS separates deterministic baselines from PT-style mixtures at a glance; it complements coverage–length rather than replacing them.

9 Discussion: where PT-like randomness hides

- **Localized CP:** a degenerate two-point scale $\hat{\sigma}(x)$ reproduces PT *exactly*; retrained $\hat{\sigma}$ varies across runs.
- **Cherry-picking:** retrain a stochastic component, report the favorable run — no task information used.
- **Randomized tie-breaking** in APS/RAPS adds mild stochasticity — worth reporting, not hiding.

Takeaways

- Coverage + length are **necessary but not sufficient**: averages reward degenerate sets.
- Ask *where a length gain comes from*; report **Interval Stability** for stochastic CP.

